


COUNTRY DOSSIER

# Senegal

Water, climate, fertilisers and crops: National study and a study of each region, potential benefits of our products, external trade

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*The Sine-Saloum delta, a UNESCO World Heritage site.*

# Foreword

This dossier describes Senegal from the angle of water, climate, fertilisers and crops. It is written for the ministries in charge of agriculture, water and trade, the regional directorates of agriculture, cooperatives and producer organisations. It answers three questions: Where is water short, for which crops, and in which regions? It then examines what our products can change, in rainfed as well as irrigated farming, and what that can mean for the country's external trade.

**Reading the sources.** Every figure refers to a source, named in the text and detailed in Annex A. Each source carries a confidence level:

**How to read this document**

- **A**: Official statistics or international database (World Bank, FAO, ECOWAS, WAEMU, Senegalese administration).
- **B**: Research institute, laboratory, United Nations agency or recognised climate model.
- **C**: Manufacturer statement, website, press or estimate.
- "M" means million; "Mm³" means million cubic metres; "t" means tonnes; "I$" means constant 2014-2016 international dollars.
- A "scenario" is a calculation based on the trials cited. It is not a measurement made in Senegal.

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# Summary

The points below summarise the dossier. The figures come from public sources: The World Bank, FAO (FAOSTAT, AQUASTAT), ECOWAS, WAEMU, Senegalese administration, PVGIS and ERA5. The production gains and water savings are scenarios, described in Annex B.

**18.93 M**
Inhabitants in 2025 (World Bank)

**1.27 M ha**
Of groundnuts, excluding shelled harvested in 2024, the main crop

**120,100 ha**
Equipped for irrigation (2024)

**+2,406 t**
Of maize per 10,000 ha treated, central scenario

**385 Mm³**
Of irrigation water saved a year, central scenario

**805 M$**
Of agricultural exports in 2024 (FAOSTAT)

**Key points**

• **A rain-fed agriculture.** 120,100 ha equipped for irrigation in 2024, for 3.91 million hectares of cropland: Almost every harvest depends on rain.

• **Groundnuts, excluding shelled: The main crop.** 1,267,592 ha harvested in 2024, for 1.68 million tonnes, or 1.33 t/ha.

• **Water is not short over the year, it is short between two rains.** National water stress is only 16.3%, but at Matam, the air demands 2,689 mm more than it rains.

• **In rainfed farming, the gain is counted in tonnes.** In the central scenario, 2,406 t of maize and 22,068 t of watermelon more per 10,000 ha treated.

• **In irrigation, the water saved enlarges the schemes.** 384.9 Mm³ a year in the central scenario, enough to irrigate about 16,535 more ha, or 13.8% of the equipped area.

• **An organic chain.** 2,780 ha of certified organic cropland in 2024: ECOFERT® is EU Bio certified.

• **Fertilisers made locally.** 54,348 t of nitrogen used in 2024; the country produces 14,039 t and imports 49,863 t.

• **Farm trade.** $805 million of exports and $2,200 million of imports in 2024. First sale: Groundnuts, shelled ($107 million). Rice weighs $497 million of purchases.

## Key figures

### Key indicators for Senegal

| Indicator | Value | Year | Source (level) |
|---|---|---|---|
| Population | 18.93 M | 2025 | World Bank (A1) |
| GDP per capita | 1,955 $ | 2025 | World Bank (A1) |
| Agriculture share of GDP | 17.1% | 2025 | World Bank (A1) |
| Agricultural employment | 29.6% | 2025 | World Bank (A1) |
| Cropland | 3,912,560 ha | 2024 | FAOSTAT (A6) |
| Area equipped for irrigation | 120,100 ha | 2024 | FAOSTAT (A6) |
| Harvested area: Groundnuts, excluding shelled | 1,267,592 ha | 2024 | FAOSTAT (A2) |
| Certified organic cropland | 2,780 ha | 2024 | FAOSTAT (A6) |
| Agricultural exports | 805 M$ | 2024 | FAOSTAT (A4) |
| Agricultural imports | 2,200 M$ | 2024 | FAOSTAT (A4) |

| Indicator | Value | Year | Source (level) |
|---|---|---|---|
| Mineral nitrogen, agricultural use | 54,348 t | 2024 | FAOSTAT (A5) |

*Source: Table based on the sources indicated.*

# 1. Green Solutions and our products

Green Solutions develops products for water, fertilisation and crop protection. Four families are of interest to this dossier. The yields cited are those published by the manufacturer or by the trials, with their confidence level.

### Products covered by this dossier

| Product | Claimed use | Published evidence (level) |
|---|---|---|
| EVERGREEN® (water-retention polymer) | Holds rain or irrigation water as a gel, near the roots. | Trials on maize, potato, watermelon, date palm and soybean. Level B to C. |
| ECOSORB® (water-retention polymer based on corn starch) | Same use; action lasting 2 to 3 years according to the website. | Soybean trial (China); laboratory acrylamide analysis. Level B to C. |
| ECOFERT® (fertiliser) | Fertiliser certified by SOHISCERT and EU Bio, usable in organic farming. | Published certificate; potato trial combined with EVERGREEN®. Level A (certificate), C (results). |
| Paulownia ALTIFOLIA® | Fast-growing tree for agroforestry. | Website figures; no trial in Senegal. Level C. |
| Water Vital®, NAPEMA® | Outside the scope of this dossier. | No independent source in the documents consulted. |

*Source: Green Solutions website (product sheets and published trials), Annex A.*

The indicative dose published on the website is 30 to 75 kg/ha.



*Groundnut crop at Djissa Aliou, in the Kaolack region.*

# 2. Senegal in figures

Senegal had 18.93 million inhabitants in 2025, 44% of them in rural areas. Agriculture accounts for 17.1% of gross domestic product and employs 30% of the labour force. The country covers 192.5 thousand km², of which 49% is agricultural land. Undernourishment affects 5.3% of the population.

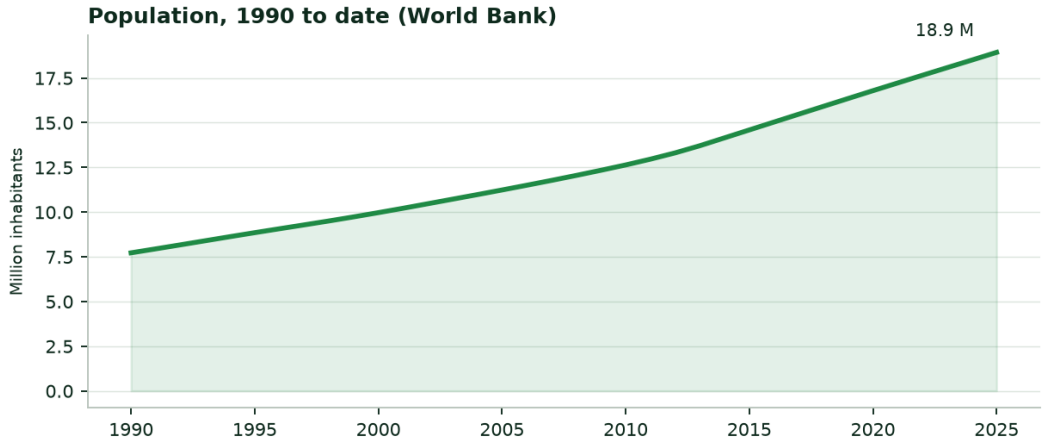



*Figure 1. Population of Senegal, 1990 to date. Source: World Bank (A1)*

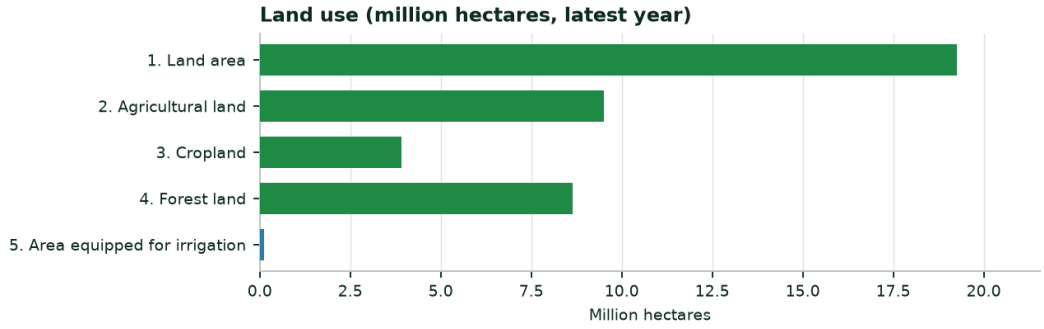



*Figure 2. Land use, latest year published. Source: FAOSTAT (A6)*

### Agricultural and water indicators, latest year published

| Indicator | Value | Year | Source |
|---|---|---|---|
| Cropland certified organic | 2,780 ha | 2024 | FAOSTAT (A6) |
| Average cereal yield | 2,004 kg/ha | 2024 | World Bank (A1) |
| Fertiliser use per hectare of arable land | 19.3 kg/ha | 2024 | World Bank (A1) |
| Water withdrawals, agricultural share | 91.3% | 2022 | World Bank (A1) |
| Internal freshwater resources | 25.8 km³ | 2022 | World Bank (A1) |
| Internal resources per inhabitant | 1,462 m³ | 2022 | World Bank (A1) |



*A Sahelian landscape in Senegal.*

# 3. Sunshine, rainfall and climate change

## 3.1 Sunshine

Global solar radiation reaches 2,183 to 2,307 kWh/m² a year depending on the region. Energy is not the limit: Water available at the right time makes the harvest.

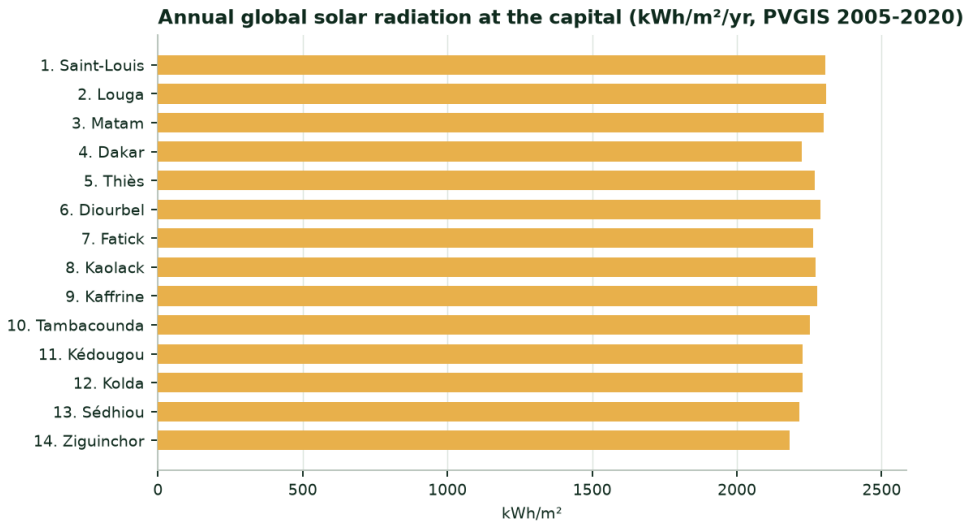



*Figure 3. Annual global solar radiation at each region's capital. Source: PVGIS (A8)*

## 3.2 Rainfall and evapotranspiration

It rains 1,158 mm a year at Ziguinchor, but only 172 mm at Saint-Louis, where reference evapotranspiration reaches 2,148 mm. Rain falls in a single season, the hivernage, from June or July to October; it decreases from south to north, and the dry season lasts seven to nine months.

**Mean monthly rainfall, 1991-2020 (mm per month)**

| City | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec |
|---|---|---|---|---|---|---|---|---|---|---|---|---|

| City | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Saint-Louis | 3 | 2 | 0 | 0 | 0 | 1 | 16 | 67 | 62 | 17 | 1 | 2 |
| Louga | 3 | 2 | 0 | 0 | 0 | 2 | 31 | 96 | 87 | 22 | 1 | 2 |
| Matam | 1 | 1 | 0 | 0 | 0 | 10 | 42 | 93 | 62 | 13 | 0 | 2 |
| Dakar | 2 | 1 | 0 | 0 | 0 | 2 | 33 | 128 | 94 | 25 | 1 | 1 |
| Thiès | 2 | 1 | 0 | 0 | 0 | 3 | 49 | 146 | 117 | 34 | 1 | 1 |
| Diourbel | 3 | 1 | 0 | 0 | 0 | 8 | 65 | 156 | 123 | 30 | 1 | 2 |
| Fatick | 2 | 1 | 0 | 0 | 0 | 8 | 80 | 196 | 148 | 45 | 1 | 2 |
| Kaolack | 2 | 1 | 0 | 0 | 0 | 12 | 88 | 204 | 150 | 40 | 1 | 2 |
| Kaffrine | 2 | 1 | 0 | 0 | 1 | 18 | 108 | 214 | 157 | 41 | 1 | 2 |
| Tambacounda | 0 | 0 | 0 | 0 | 3 | 35 | 130 | 206 | 163 | 41 | 2 | 1 |
| Kédougou | 0 | 0 | 1 | 1 | 17 | 76 | 164 | 240 | 226 | 68 | 6 | 0 |
| Kolda | 0 | 0 | 0 | 0 | 3 | 46 | 203 | 308 | 214 | 67 | 5 | 1 |
| Sédhiou | 0 | 1 | 0 | 0 | 2 | 47 | 216 | 342 | 238 | 76 | 6 | 2 |
| Ziguinchor | 1 | 1 | 0 | 0 | 2 | 51 | 268 | 389 | 302 | 128 | 14 | 1 |

*Source: ERA5 via Open-Meteo (A9); author's calculation.*

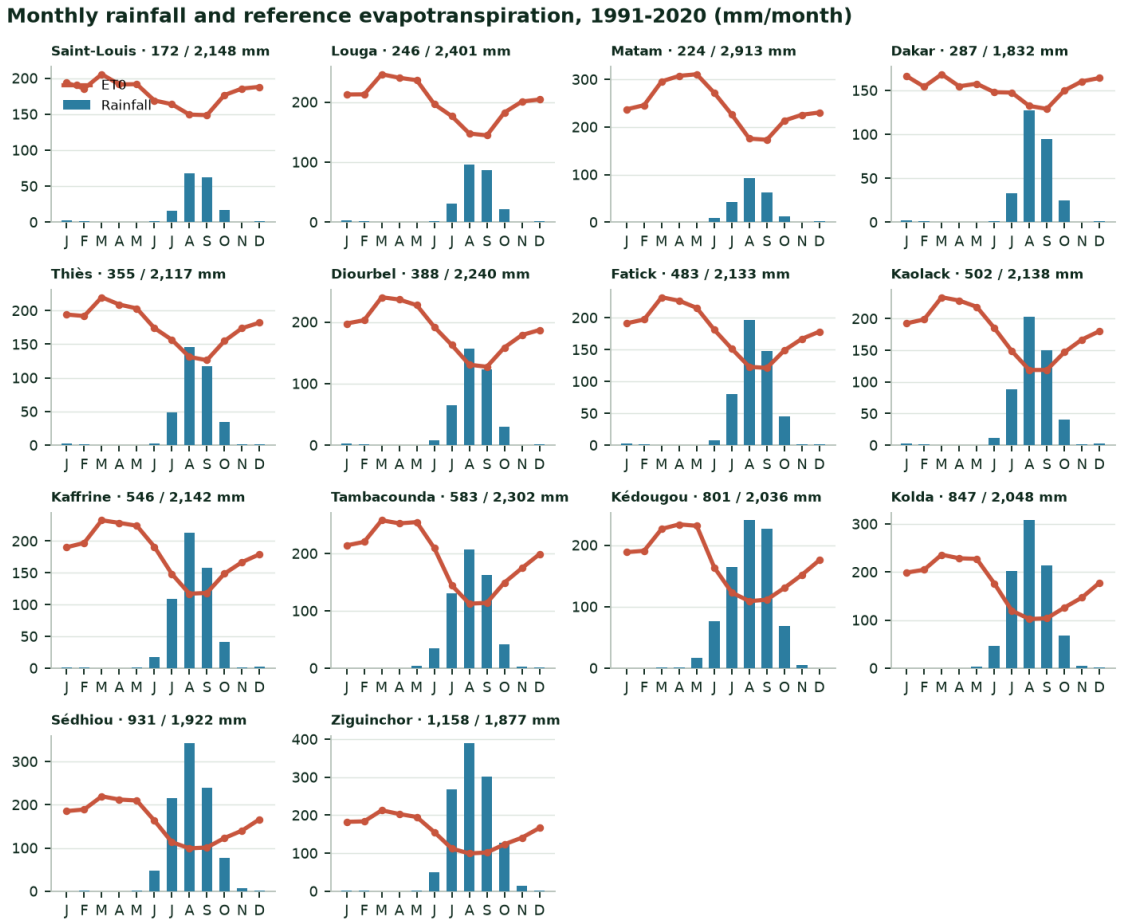



*Figure 4. Monthly rainfall and reference evapotranspiration by city, 1991-2020. Source: ERA5 via Open-Meteo (A9)*

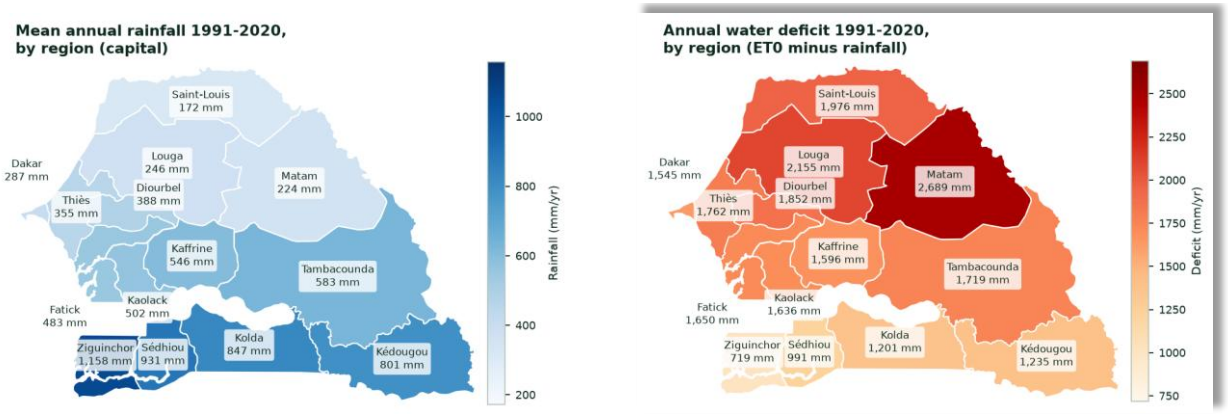



*Figure 5. Mean annual rainfall 1991-2020, by region. Source: ERA5 Annual water deficit, by region. Source: ERA5 (A9), Natural Earth (A10). (A9), Natural Earth (A10).*

## 3.3 Climate change

Mean annual temperature is rising. At Saint-Louis it went from 25.2 °C in 1981 to 26.5 °C in 2024, a trend of 0.27 °C per decade. At Ziguinchor the trend is 0.09 °C per decade, from 26.9 °C to 27.6 °C. Warmer air dries the soil faster between two rains.

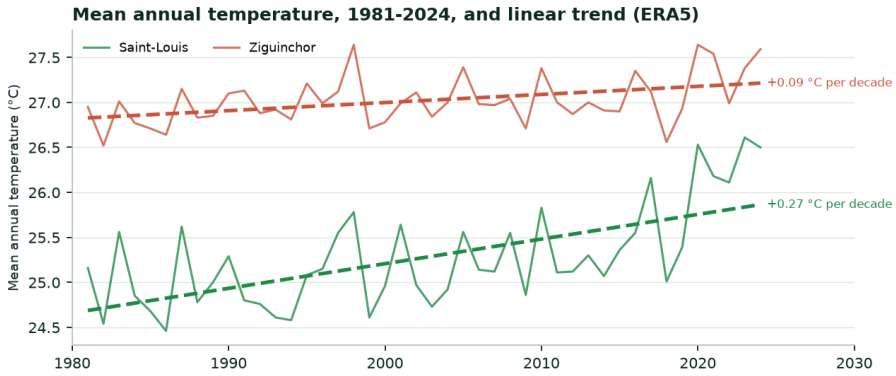



*Figure 6. Mean annual temperature, 1981-2024, and linear trend. Source: ERA5 (A9)*



*A baobab near Darou Khoudoss, in the Thiès region.*

# 4. Water: Resources, irrigated areas, withdrawals

## 4.1 Resources and withdrawals

Total freshwater withdrawals are 3,062 million m³ a year (2022). Agriculture withdraws 91.3% of it, that is 2,796 Mm³. Internal resources are 25.8 km³ a year and water stress is 16.3%: The country has water, but little of it is mobilised for crops.

**Freshwater: Withdrawals and resources**

| Indicator | Value | Year | Source |
|---|---|---|---|
| Total withdrawals | 3,062 Mm³ | 2022 | World Bank (A1) |
| Agricultural share | 91.3% | 2022 | World Bank (A1) |
| Agricultural withdrawals | 2,795.7 Mm³ | 2022 | Calculated from A1 |
| Area equipped for irrigation | 120,100 ha | 2024 | FAOSTAT (A6) |
| Agricultural withdrawals per equipped hectare | 23,278 m³/ha | — | Calculation (Annex B) |
| Water stress | 16.3% | 2022 | World Bank (A1) |

## 4.2 Irrigated areas

FAOSTAT counts 120,100 hectares equipped for irrigation in 2024. FAO uses an irrigation potential of 409,000 ha; depending on the study, it ranges from 160,000 to more than 640,000 ha (A11). Irrigation mostly serves rice and tomatoes in the Senegal river valley, Richard-Toll sugar cane, onions and market gardening in the Niayes.



*A pumping station in the Senegal river valley.*

## 4.3 What the water saving represents

In the central scenario, the saving is 384.9 million m³ a year, or 13.8% of agricultural withdrawals. At 23,278 m³ per equipped hectare, this water would irrigate about 16,535 more hectares, or 13.8% of today's equipped area.

# 5. Fertilisers: Quantities, purchases, spending

## 5.1 Quantities used

In 2024, agricultural use reaches 54,348 tonnes of nitrogen, 11,180 tonnes of phosphate (P2O5) and 8,346 tonnes of potash (K2O). The country produces 14,039 tonnes of nitrogen and 7,367 tonnes of phosphate (P2O5); it imported 49,863 tonnes of nitrogen in the same year. Per hectare of arable land, use is 19.3 kg per hectare.

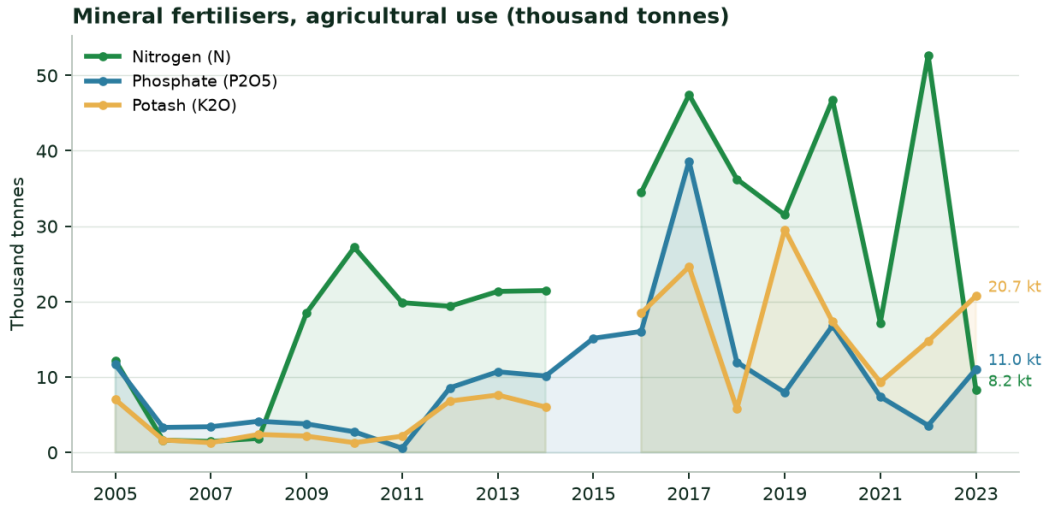



*Figure 7. Mineral fertilisers, agricultural use, 2005-2023. Source: FAOSTAT (A5)*

## 5.2 Regulatory framework

ECOWAS Regulation C/REG.13/12/12 governs fertiliser quality control in the West African area: Registration before sale, labelling, control (A13).



*A woman carrying a sack of onions.*

## 5.3 Indicative spending

In 2024, the indicative minimum spending on mineral fertilisers reaches $56 million, at international fertiliser prices (calculation in Annex B, lower bound).

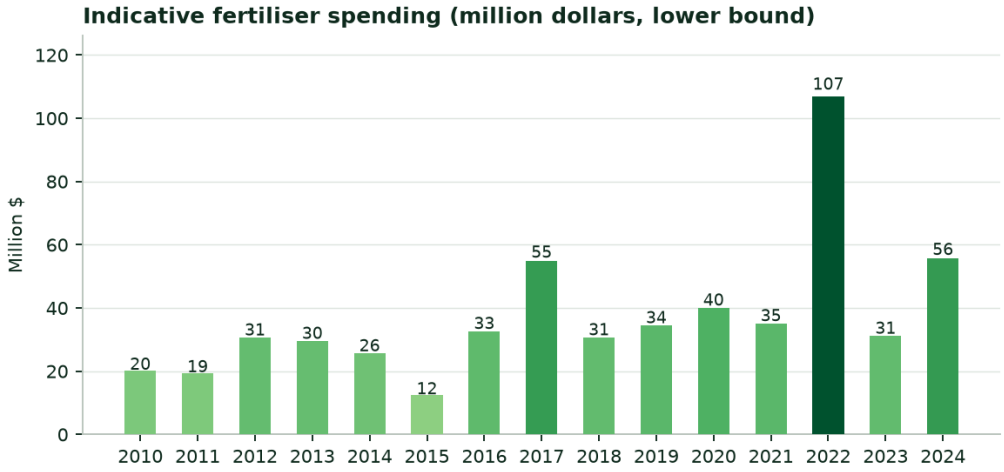



*Figure 8. Indicative fertiliser spending, lower bound. Source: Author's calculation from A5 and A7*

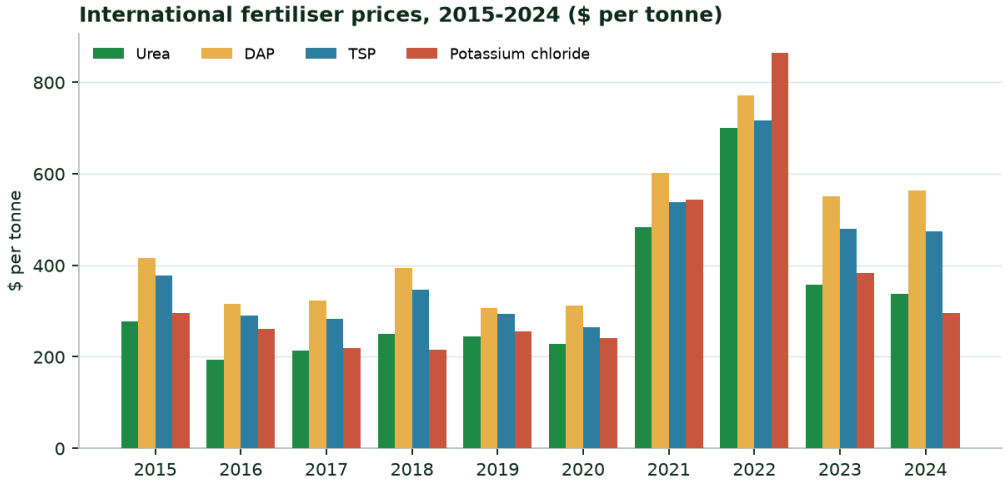



*Figure 9. International prices of urea, DAP, TSP and potassium chloride. Source: World Bank, Pink Sheet (A7)*

# 6. The regions at a glance

The table compares the country's fourteen regions using their capital. Climate figures are 1991-2020 averages, except radiation (2005-2020).

**Climate of the capitals, 1991-2020 means**

| Region (city) | Rainfall (mm/yr) | ET0 (mm/yr) | Deficit (mm/yr) | Radiation (kWh/m²/yr) | Temperature (°C) |
|---|---|---|---|---|---|
| Saint-Louis (Saint-Louis) | 172 | 2,148 | 1,976 | 2,304 | 25.2 |
| Louga (Louga) | 246 | 2,401 | 2,155 | 2,307 | 26.7 |
| Matam (Matam) | 224 | 2,913 | 2,689 | 2,299 | 30.4 |
| Dakar (Dakar) | 287 | 1,832 | 1,545 | 2,224 | 24.6 |
| Thiès (Thiès) | 355 | 2,117 | 1,762 | 2,269 | 25.5 |
| Diourbel (Diourbel) | 388 | 2,240 | 1,852 | 2,289 | 26.9 |
| Fatick (Fatick) | 483 | 2,133 | 1,650 | 2,263 | 26.7 |
| Kaolack (Kaolack) | 502 | 2,138 | 1,636 | 2,272 | 27.1 |
| Kaffrine (Kaffrine) | 546 | 2,142 | 1,596 | 2,277 | 27.9 |
| Tambacounda (Tambacounda) | 583 | 2,302 | 1,719 | 2,253 | 29.5 |
| Kédougou (Kédougou) | 801 | 2,036 | 1,235 | 2,228 | 28.6 |

| Region (city) | Rainfall (mm/yr) | ET0 (mm/yr) | Deficit (mm/yr) | Radiation (kWh/m²/yr) | Temperature (°C) |
|---|---|---|---|---|---|
| Kolda (Kolda) | 847 | 2,048 | 1,201 | 2,226 | 28.1 |
| Sédhiou (Sédhiou) | 931 | 1,922 | 991 | 2,215 | 27.3 |
| Ziguinchor (Ziguinchor) | 1,158 | 1,877 | 719 | 2,183 | 27.0 |

*Source: ERA5 via Open-Meteo (A9); PVGIS (A8); author's calculation.*

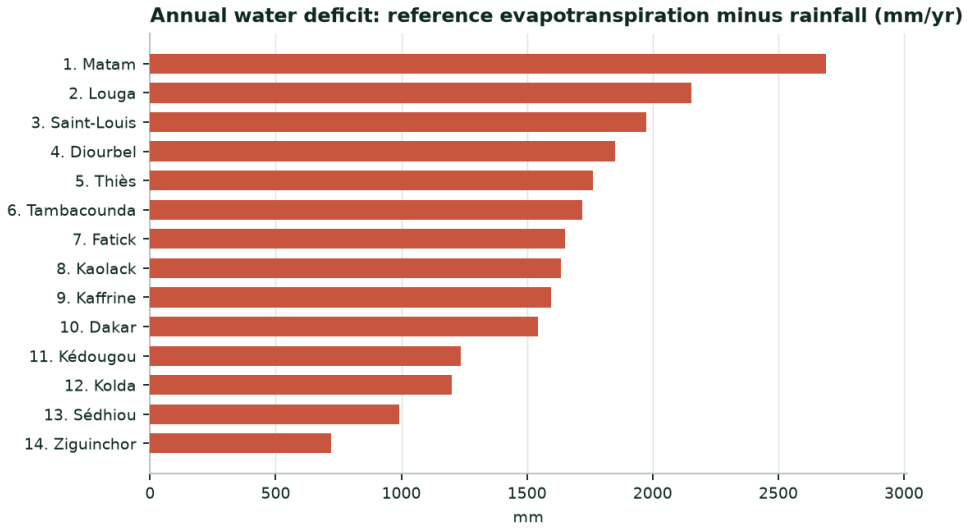



*Figure 10. Annual water deficit: Reference evapotranspiration minus rainfall. Source: ERA5 (A9)*



*Niokolo-Koba National Park, in the east of the country.*

# 7. The twenty main crops: Three rankings

## 7.1 By harvested area

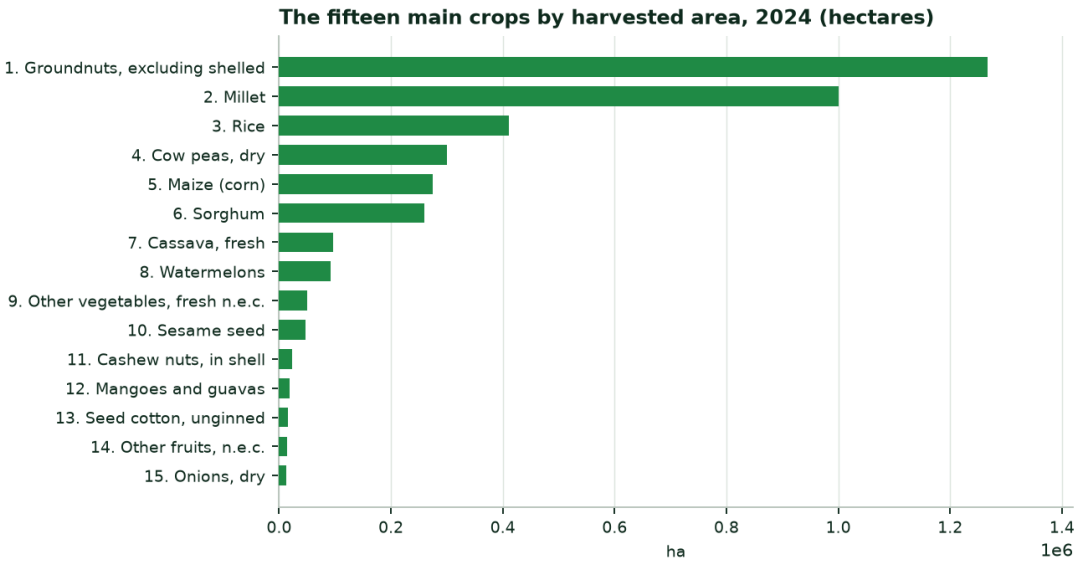



*Figure 11. The fifteen main crops by harvested area, 2024 (hectares). Source: FAOSTAT (A2)*

### The ten main crops by area, 2024

| Crop | Harvested area (ha) | Production (t) | Yield (t/ha) |
|---|---|---|---|
| Groundnuts, excluding shelled | 1,267,592 | 1,681,953 | 1.33 |
| Millet | 1,000,000 | 1,106,000 | 1.11 |
| Rice | 410,271 | 1,580,000 | 3.85 |
| Cow peas, dry | 300,395 | 212,909 | 0.71 |
| Maize (corn) | 275,000 | 774,000 | 2.81 |
| Sorghum | 260,000 | 445,000 | 1.71 |
| Cassava, fresh | 96,844 | 1,433,575 | 14.80 |
| Watermelons | 92,607 | 1,362,440 | 14.71 |
| Other vegetables, fresh n.e.c. | 49,829 | 212,695 | 4.27 |
| Sesame seed | 47,000 | 45,000 | 0.96 |

*Source: FAOSTAT (A2).*

## 7.2 By production

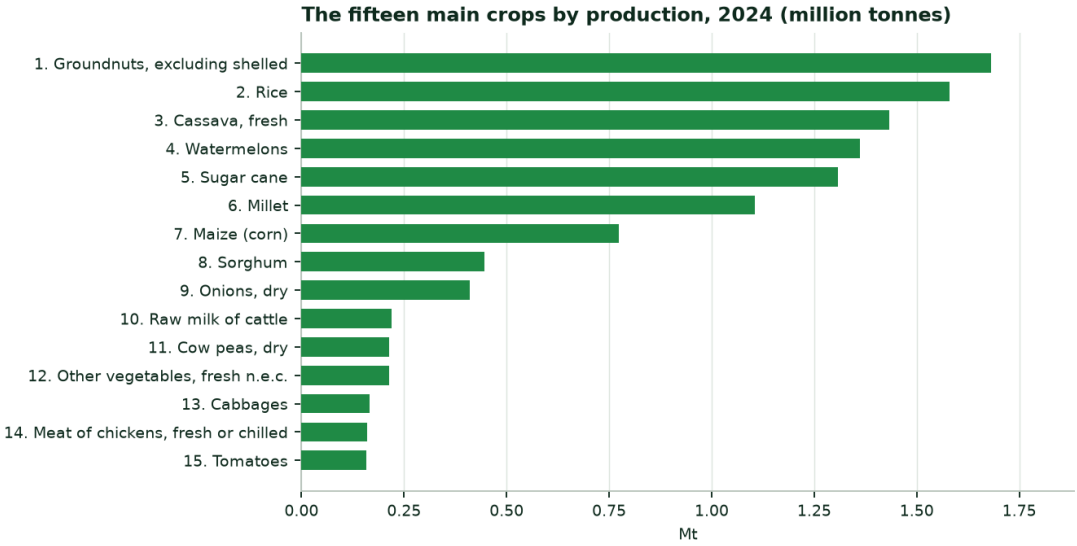



*Figure 12. The fifteen main crops by production, 2024 (million tonnes). Source: FAOSTAT (A2)*

## 7.3 By production value

By production value, the three main crops are: Groundnuts, excluding shelled, rice and millet (FAOSTAT, 2023, constant international dollars).

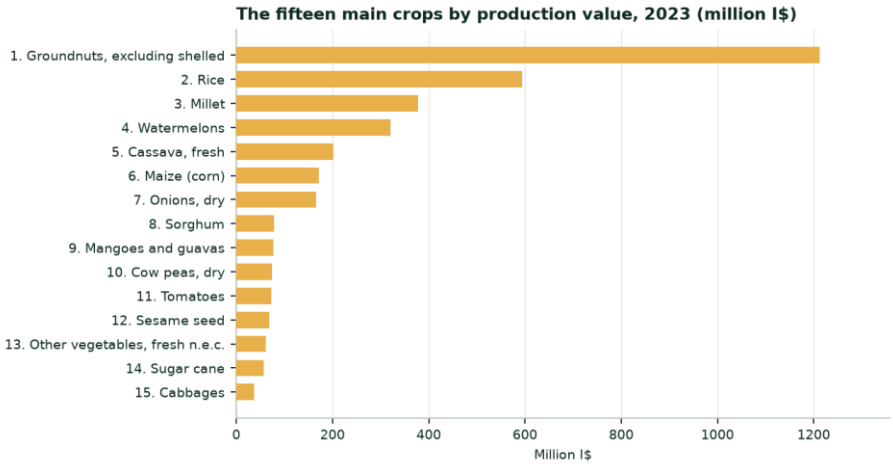



*Figure 13. The fifteen main crops by production value, 2023 (million I$). Source: FAOSTAT (A3)*



*Tomato plants in Senegal.*

# 8. Extrapolation by crop: Irrigation and rainfed crops

## 8.1 Method

**Method in brief**

• **Irrigation.** National agricultural water is shared between irrigated crops (rice, vegetables, potato and sugar cane), in proportion to their harvested areas; the equipped area is shared the same way.

• **Rainfed crops.** For crops that have a trial (maize and watermelon), the production gain is calculated per 10,000 ha treated, at the national average 2024 yield.

• Each calculation applies the effect of the trial cited: A quarter (cautious), half (central), all of it (high). Without a trial: 5, 10 and 20% of water, with no yield gain.

## 8.2 Trials used as analogues

**Trials cited and their use in the calculation**

| Trial | Crop | Water | Yield | Use in the calculation |
|---|---|---|---|---|
| Maize, Türkiye (TIGEM, Konya, 2025) | Maize | -38.9% | +17.1% | Maize: Yield |
| Soybean, China | Soybean | Assumption | +11.3% | Not used; cited for reference |
| Watermelon, Mexico (2023) | Watermelon | -60.5% | +30.0% | Watermelon: Yield |
| Potato, South Africa (2025) | Potato | -52.0% | +31.4% | Potato: Water and yield |
| Date palm, UAE (ICBA, 2021) | Palm | -82.0% | +20.0% | Not used; cited for reference |

*Source: A14 to A18 (published trials and manufacturer's website, levels B and C).*

No trial has yet been conducted in Senegal: The figures below apply the trials cited to the country's average yields.

## 8.3 Rainfed crops: Maize and Watermelon

Average 2024 yields (FAOSTAT): Maize 2.81 t/ha and watermelon 14.71 t/ha. A water-retention polymer keeps the water of each rain near the roots and helps the crop through dry days during the season.

### Extra production per 10,000 ha treated (tonnes a year)

| Crop | Average yield (t/ha) | Cautious | Central | High |
|---|---|---|---|---|
| Maize | 2.81 | 1,203 | 2,406 | 4,813 |
| Watermelon | 14.71 | 11,034 | 22,068 | 44,136 |

*Source: FAOSTAT (A2); trials A14 to A18; author's calculation (Annex B).*

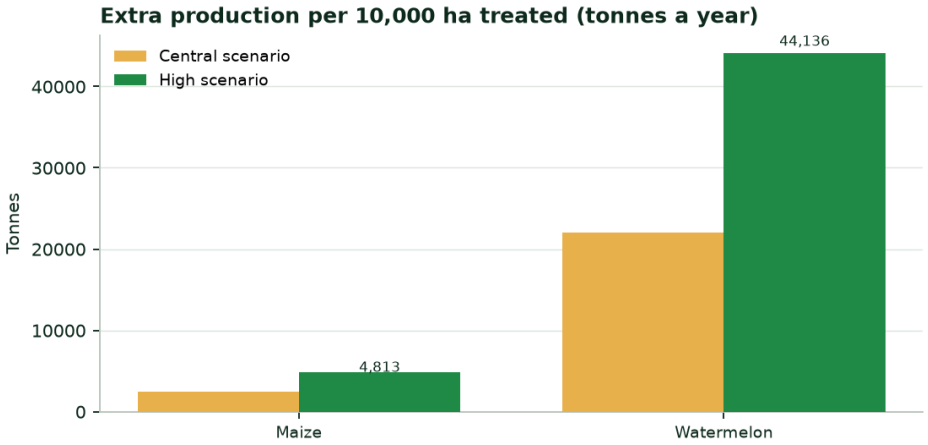



*Figure 14. Extra production per 10,000 ha treated, central and high scenarios. Source: Author's calculation*

## 8.4 Irrigation: Rice, Vegetables, Potato and Sugar cane

### Equipped area assigned, trial used and water saved (central scenario)

| Crop group | Equipped area (ha) | Share | Trial or assumption | Water saved (Mm³/yr) |
|---|---|---|---|---|
| Rice (paddy) | 94,810 | 78.9% | Generic assumption, no trial on rice | 220.7 |
| Vegetables (onion, tomato, cabbage, carrot, others) | 21,234 | 17.7% | Watermelon, Mexico (2023) | 149.5 |
| Potato | 1,406 | 1.2% | Potato, South Africa (2025) | 8.5 |
| Sugar cane | 2,650 | 2.2% | Generic assumption, no trial | 6.2 |

*Source: Author's calculation (Annex B).*

### Water saved, extra irrigable hectares and production gain, by scenario

| Scenario | Water (Mm³/yr) | Extra hectares | Production (t/yr) |
|---|---|---|---|
| Cautious | 192.4 | 8,267 | 21,310 |
| Central | 384.9 | 16,535 | 42,621 |
| High | 769.8 | 33,069 | 85,241 |

*Source: Author's calculation (Annex B).*

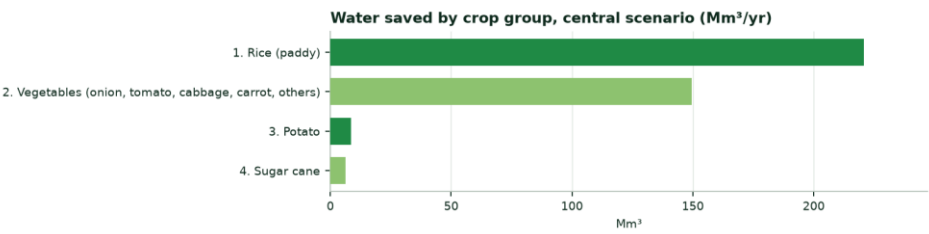



*Figure 15. Water saved by crop group, central scenario. Source: Author's calculation*

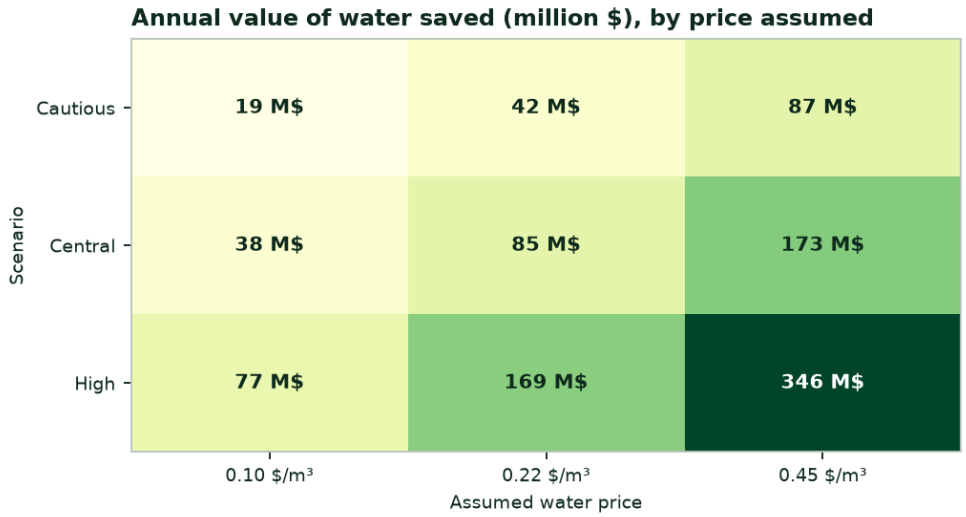



*Figure 16. Annual value of the water saved, by price assumed per cubic metre. Source: Author's calculation; prices assumed, no national tariff*

## 8.5 Reading the results

**Key points**

• In Senegal, the main stake is rainfed farming: Only 120,100 ha are equipped for irrigation, for 3,912,560 ha of cropland.

• The "rice" group gives 57% of the irrigation water saving (generic assumption, no trial on rice).

• The gains are counted by scenario and, in rainfed farming, per 10,000 ha treated: A cooperative or a public programme can scale them to its own area.

# 9. Potential benefits of our products

The table lists what the products may bring according to the documents available, with their confidence level and their limit.

**Potential benefits, by situation**

| Benefit | Published figures | Source (level) | Limit |
|---|---|---|---|
| Irrigation water saving | Date palm −82%; watermelon −60.5%; potato −52%; maize −38.9% | A14 to A18 (B, C) | One trial per crop; none in Senegal |
| Yield | Maize +17.1%; potato +31.4%; watermelon +30%; date palm +20%; soybean +11.2 to +11.4% | A14 to A18 (B, C) | Trials conducted under irrigation or in other climates |
| Organic farming (ECOFERT®) | Certified by SOHISCERT and EU Bio | A21 (A) | Valid for the certified product |
| Safety: Acrylamide (ECOSORB®) | Below 3.0 µg/g | A19 (B) | One laboratory, one date |
| REACH compliance (EVERGREEN®) | ≤ 0.1% of substances of very high concern in 233 substances tested | A20 (B) | Screening, not an approval |
| Agroforestry (Paulownia ALTIFOLIA®) | 13 to 15 m in 6 to 8 years (website) | A22 (C) | No trial in Senegal |

*Source: Annex A.*



*Market gardens at Lake Mbawane, Kayar.*

# 10. External trade: Exports, imports, importing

## 10.1 Overview, 2023 and 2024

**Agricultural products and total merchandise trade (million $)**

| Indicator | 2023 | 2024 |
|---|---|---|
| Agricultural exports (crops and livestock) | 827 | 805 |
| Agricultural imports (crops and livestock) | 2,224 | 2,200 |
| Exports, all merchandise | 5,313 | 6,408 |
| Imports, all merchandise | 11,064 | 11,807 |

*Source: FAOSTAT (A4).*

## 10.2 Exports

The three main farm products exported in 2024 are: Groundnuts, shelled ($107 million), cashew nuts, in shell ($86 million) and cigarettes ($56 million).

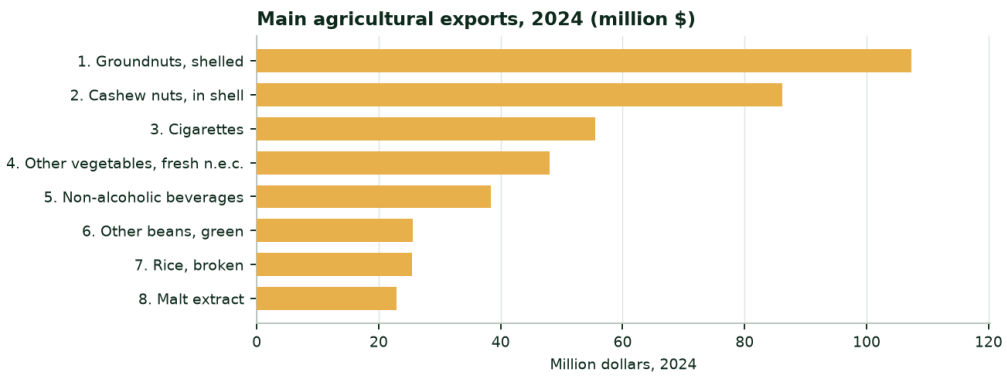



*Figure 17. Main agricultural exports, 2024. Source: FAOSTAT (A4)*

## 10.3 Imports

The main farm purchases are: Wheat ($282 million), malt extract ($216 million) and maize (corn) ($128 million). Rice weighs $497 million (husked, broken and milled rice): Every tonne grown locally reduces this bill.

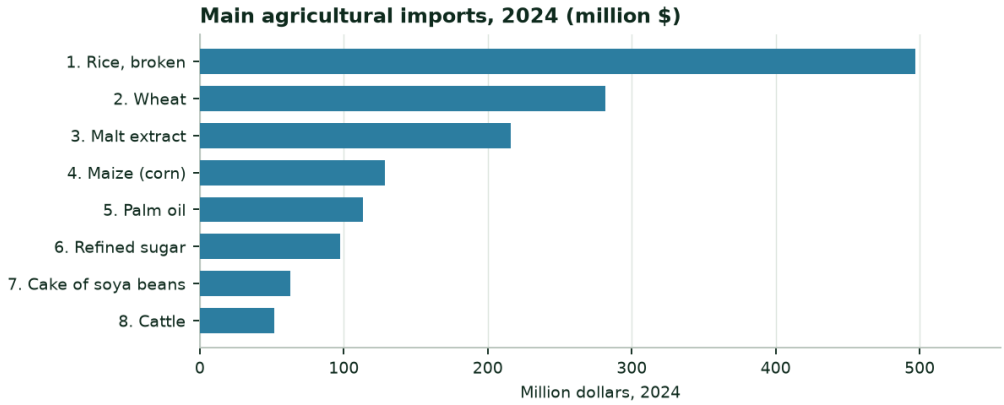



*Figure 18. Main agricultural imports, 2024. Source: FAOSTAT (A4)*



*Figure 19. Mangoes from Senegal.*



*Senegalese tomatoes at the market.*

## 10.4 Importing water-retention polymers into Senegal

**What to know**

- **Customs duty (code 3906.90.00.00, acrylic polymers in primary forms):** 5% under the ECOWAS common external tariff (A12).
- **Levies:** 1% statistical fee, 1% WAEMU community solidarity levy and 0.5% ECOWAS community levy (A12).
- **VAT:** 18% (A12).
- **Registration:** A fertiliser must be registered before sale, under ECOWAS Regulation C/REG.13/12/12 (A13).
- **Classification:** Code 3906.90 is an inference for our products; Senegalese customs settle it.

# 11. A study of each region

Each region is described through the city that represents it. The figures describe the climate; the water deficit (evapotranspiration minus rainfall) shows where water is most short. The description of crops is qualitative.



*Sugar cane rows at Richard-Toll.*

## Saint-Louis (Saint-Louis)

Senegal river valley: Irrigated rice, processing tomato, onion and Richard-Toll sugar cane.

| Indicator | Value |
|---|---|
| Annual rainfall, 1991-2020 | 172 mm |
| Reference evapotranspiration | 2,148 mm |
| Annual water deficit | 1,976 mm |
| Months with less than 50 mm of rain | 10 |
| Global solar radiation | 2,304 kWh/m² |
| Mean temperature | 25.2 °C |

*Source: ERA5 via Open-Meteo (A9); PVGIS (A8); author's calculation.*

## Louga (Louga)

Livestock country and the northern Niayes: Groundnuts, cowpea, watermelon and market gardening.

| Indicator | Value |
|---|---|
| Annual rainfall, 1991-2020 | 246 mm |
| Reference evapotranspiration | 2,401 mm |
| Annual water deficit | 2,155 mm |
| Months with less than 50 mm of rain | 10 |
| Global solar radiation | 2,307 kWh/m² |
| Mean temperature | 26.7 °C |

*Source: ERA5 via Open-Meteo (A9); PVGIS (A8); author's calculation.*

## Matam (Matam)

Middle river valley: Irrigated rice, flood-recession crops, millet and livestock.

| Indicator | Value |
|---|---|
| Annual rainfall, 1991-2020 | 224 mm |
| Reference evapotranspiration | 2,913 mm |
| Annual water deficit | 2,689 mm |
| Months with less than 50 mm of rain | 10 |
| Global solar radiation | 2,299 kWh/m² |
| Mean temperature | 30.4 °C |

*Source: ERA5 via Open-Meteo (A9); PVGIS (A8); author's calculation.*

## Dakar (Dakar)

Peri-urban market gardening in the Niayes and poultry, for the country's largest market.

| Indicator | Value |
|---|---|
| Annual rainfall, 1991-2020 | 287 mm |
| Reference evapotranspiration | 1,832 mm |
| Annual water deficit | 1,545 mm |
| Months with less than 50 mm of rain | 10 |
| Global solar radiation | 2,224 kWh/m² |
| Mean temperature | 24.6 °C |

*Source: ERA5 via Open-Meteo (A9); PVGIS (A8); author's calculation.*

## Thiès (Thiès)

Niayes and the western groundnut basin: Market gardening, mango, groundnuts, millet and cowpea.

| Indicator | Value |
|---|---|
| Annual rainfall, 1991-2020 | 355 mm |
| Reference evapotranspiration | 2,117 mm |
| Annual water deficit | 1,762 mm |
| Months with less than 50 mm of rain | 10 |
| Global solar radiation | 2,269 kWh/m² |
| Mean temperature | 25.5 °C |

*Source: ERA5 via Open-Meteo (A9); PVGIS (A8); author's calculation.*

## Diourbel (Diourbel)

The heart of the groundnut basin: Groundnuts, millet and cowpea on sandy soils.

| Indicator | Value |
|---|---|
| Annual rainfall, 1991-2020 | 388 mm |
| Reference evapotranspiration | 2,240 mm |
| Annual water deficit | 1,852 mm |
| Months with less than 50 mm of rain | 9 |
| Global solar radiation | 2,289 kWh/m² |
| Mean temperature | 26.9 °C |

*Source: ERA5 via Open-Meteo (A9); PVGIS (A8); author's calculation.*

## Fatick (Fatick)

Sine and Saloum: Groundnuts and millet, on land partly affected by salt.

| Indicator | Value |
|---|---|
| Annual rainfall, 1991-2020 | 483 mm |
| Reference evapotranspiration | 2,133 mm |
| Annual water deficit | 1,650 mm |
| Months with less than 50 mm of rain | 9 |
| Global solar radiation | 2,263 kWh/m² |
| Mean temperature | 26.7 °C |

*Source: ERA5 via Open-Meteo (A9); PVGIS (A8); author's calculation.*

## Kaolack (Kaolack)

Groundnut basin: Groundnuts, millet and maize; a major farm market.

| Indicator | Value |
|---|---|
| Annual rainfall, 1991-2020 | 502 mm |
| Reference evapotranspiration | 2,138 mm |
| Annual water deficit | 1,636 mm |
| Months with less than 50 mm of rain | 9 |
| Global solar radiation | 2,272 kWh/m² |
| Mean temperature | 27.1 °C |

*Source: ERA5 via Open-Meteo (A9); PVGIS (A8); author's calculation.*

## Kaffrine (Kaffrine)

The eastern groundnut basin: Groundnuts, millet, maize and sorghum.

| Indicator | Value |
|---|---|
| Annual rainfall, 1991-2020 | 546 mm |
| Reference evapotranspiration | 2,142 mm |
| Annual water deficit | 1,596 mm |
| Months with less than 50 mm of rain | 9 |
| Global solar radiation | 2,277 kWh/m² |
| Mean temperature | 27.9 °C |

*Source: ERA5 via Open-Meteo (A9); PVGIS (A8); author's calculation.*

## Tambacounda (Tambacounda)

The east of the country: Cotton, millet, maize, groundnuts, and bananas along the Gambia river.

| Indicator | Value |
|---|---|
| Annual rainfall, 1991-2020 | 583 mm |
| Reference evapotranspiration | 2,302 mm |
| Annual water deficit | 1,719 mm |
| Months with less than 50 mm of rain | 9 |
| Global solar radiation | 2,253 kWh/m² |
| Mean temperature | 29.5 °C |

*Source: ERA5 via Open-Meteo (A9); PVGIS (A8); author's calculation.*

## Kédougou (Kédougou)

South-eastern uplands: Maize, fonio, sorghum and cotton.

| Indicator | Value |
|---|---|
| Annual rainfall, 1991-2020 | 801 mm |
| Reference evapotranspiration | 2,036 mm |
| Annual water deficit | 1,235 mm |
| Months with less than 50 mm of rain | 7 |
| Global solar radiation | 2,228 kWh/m² |
| Mean temperature | 28.6 °C |

*Source: ERA5 via Open-Meteo (A9); PVGIS (A8); author's calculation.*

## Kolda (Kolda)

Upper Casamance: Cotton, maize, rice, groundnuts and cashew.

| Indicator | Value |
|---|---|
| Annual rainfall, 1991-2020 | 847 mm |
| Reference evapotranspiration | 2,048 mm |
| Annual water deficit | 1,201 mm |
| Months with less than 50 mm of rain | 8 |
| Global solar radiation | 2,226 kWh/m² |
| Mean temperature | 28.1 °C |

*Source: ERA5 via Open-Meteo (A9); PVGIS (A8); author's calculation.*

## Sédhiou (Sédhiou)

Middle Casamance: Rice, cashew, groundnuts, maize and mango.

| Indicator | Value |
|---|---|
| Annual rainfall, 1991-2020 | 931 mm |
| Reference evapotranspiration | 1,922 mm |
| Annual water deficit | 991 mm |
| Months with less than 50 mm of rain | 8 |
| Global solar radiation | 2,215 kWh/m² |
| Mean temperature | 27.3 °C |

*Source: ERA5 via Open-Meteo (A9); PVGIS (A8); author's calculation.*

## Ziguinchor (Ziguinchor)

Lower Casamance: Lowland and mangrove rice, cashew, mango and citrus.

| Indicator | Value |
|---|---|
| Annual rainfall, 1991-2020 | 1,158 mm |
| Reference evapotranspiration | 1,877 mm |
| Annual water deficit | 719 mm |
| Months with less than 50 mm of rain | 7 |
| Global solar radiation | 2,183 kWh/m² |
| Mean temperature | 27.0 °C |

*Source: ERA5 via Open-Meteo (A9); PVGIS (A8); author's calculation.*

# Annex A. Sources and confidence levels

| Ref. | Source | Level |
|---|---|---|
| A1 | World Bank, World Development Indicators (WDI), Senegal | A |
| A2 | FAOSTAT, crops and livestock (QCL), areas, production, 2014-2024 | A |
| A3 | FAOSTAT, value of production (QV), constant 2014-2016 I$, 2019-2024 | A |
| A4 | FAOSTAT, trade in crops and livestock products, 2018-2024 | A |
| A5 | FAOSTAT, mineral fertilisers, agricultural use, production and imports, 2005-2023 | A |
| A6 | FAOSTAT, land use, areas equipped for irrigation, organic farming | A |
| A7 | World Bank, Pink Sheet (world prices of urea, DAP, TSP and potassium chloride) | A |
| A8 | PVGIS (European Commission Joint Research Centre), radiation, 2005-2020 | B |
| A9 | ERA5 (Copernicus), via Open-Meteo: Rainfall, reference evapotranspiration, temperature, 1981-2024 | B |
| A10 | Natural Earth, first-level administrative boundaries and country outline | B |
| A11 | FAO, AQUASTAT, Senegal country profile (2005 version): Irrigation potential and range of estimates | A |
| A12 | ECOWAS common external tariff (heading 3906.90.00.00: Customs duty, statistical fee, levies); Senegal general tax code (VAT) | A |
| A13 | ECOWAS Regulation C/REG.13/12/12 on fertiliser quality control | A |
| A14 | Soybean trial, China (yield), manufacturer's website | C |
| A15 | Date palm trial, UAE (ICBA, 2021), manufacturer's website | B |
| A16 | Watermelon trial, Mexico (2023), manufacturer's website | C |
| A17 | Potato trial, South Africa (2025), manufacturer's website | C |
| A18 | Maize trial, Türkiye (TIGEM, Konya, 2025), manufacturer's website | B |
| A19 | ECOSORB® acrylamide analysis (Polymex laboratory, 2024) | B |
| A20 | EVERGREEN® REACH / SVHC screening (SGS, 2023) | B |
| A21 | ECOFERT® certificate (SOHISCERT, EU Bio) | A |
| A22 | Paulownia ALTIFOLIA® sheet figures (website) | C |

# Annex B. Calculation method

**1. Agricultural water.** Total withdrawals (3,062 Mm³, 2022) × agricultural share (91.31%) = 2,795.7 Mm³.

**2. Irrigation.** The equipped area (120,100 ha) and agricultural water are shared between irrigated groups (rice, vegetables, potato and sugar cane) in proportion to their harvested areas (519,707 ha), without exceeding each group's harvested area. Our sources have no irrigation statistic by crop. Other crops are treated as rainfed and receive no irrigation water in the calculation. Groundnuts, millet, sorghum, cowpea and mango are treated as rainfed.

**3. Scenarios.** Water saved = group water × trial effect: A quarter (cautious), half (central), all of it (high). Without a trial: 5, 10 and 20%.

**4. Extra hectares.** Water saved ÷ water per equipped hectare (23,278 m³/ha).

**5. Rainfed crops.** Gain per 10,000 ha = 10,000 × average 2024 yield (production ÷ harvested area, FAOSTAT) × trial effect on yield (maize 17.1% and watermelon 30.0%), with the same quarter, half and whole.

**6. Value of water.** Three assumed prices (0.10; 0.22 and 0.45 $/m³); no national tariff was adopted.

**7. Fertilisers.** Minimum spending = nitrogen ÷ 0.46 × urea price + phosphate ÷ 0.46 × TSP price + potash ÷ 0.60 × potassium chloride price (A7).

**8. Climate.** Rainfall, reference evapotranspiration (FAO-56) and temperature: 1991-2020 means of the ERA5 grid cell containing each city; radiation: PVGIS, 2005-2020. Temperature trends are linear regressions over 1981-2024. ERA5 is a reanalysis: Its rainfall can differ from local station records. On the maps, each region takes the value of its capital.

**9. Trade.** Totals are those of the FAOSTAT crops and livestock line; products are listed one by one, without aggregates. Rice amounts add husked rice, broken rice and milled rice.

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*The Sine-Saloum delta.*

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